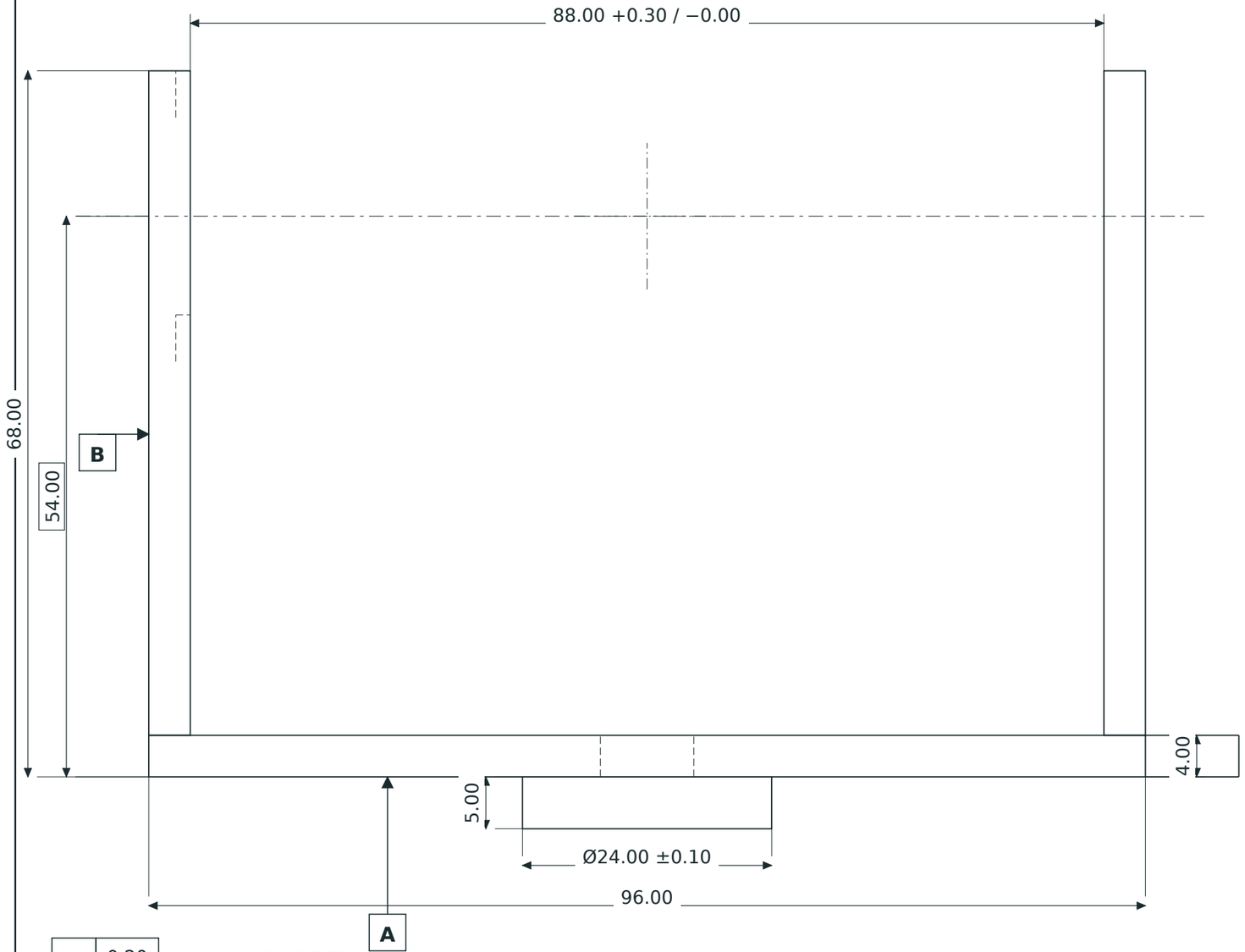
NODI / MECHANICAL DESIGN

PROTOTYPE DEFINITION — NOT A PRODUCTION RELEASE

FRONT / X–Z

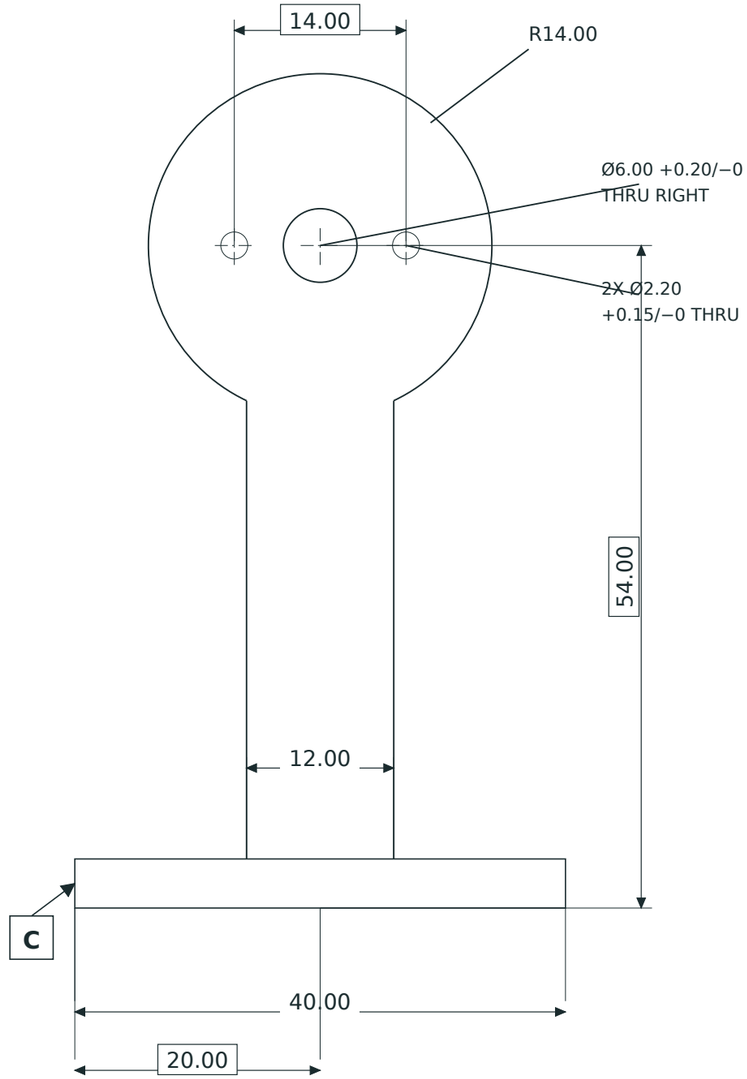


▱	0.20

APPLIES TO A (annular bottom plate face; exclude pilot)

Front projection: pads have R14 outline in Y–Z view. Front view does not define pad profile.

RIGHT / Y–Z



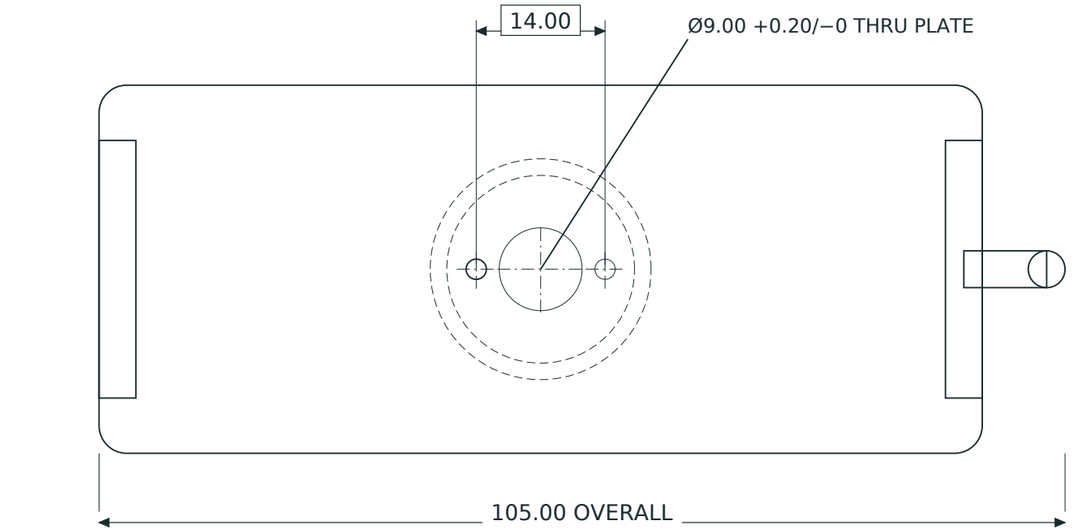
LEFT BORE: Ø4.20 +0.10/−0 THRU (opposite arm)

⌖	Ø0.30	A	B	C

Applies separately to LEFT Ø4.2 and RIGHT Ø6 axes.
Hole Y=0, Z=88 share basic axis; horn holes at Y=±7.

NODI-P03 PAN–TILT YOKE MATERIAL: PETG, FFF + FINISH DRILL / PROCESS NOTES ON SHEET 2	**REV A / 2026-10-24** Scale: 2:1 views (A3)	**David Borsky** AI-assisted design study
UNITS: mm | THIRD ANGLE PROJECTION | DO NOT SCALE DRAWING General ±0.25 linear / ±1° angular unless limits, basic dimensions or GD&T apply.	ASME Y14.5-2018 principles Unbuilt / inspect before assembly	**SHEET 1 OF 2**

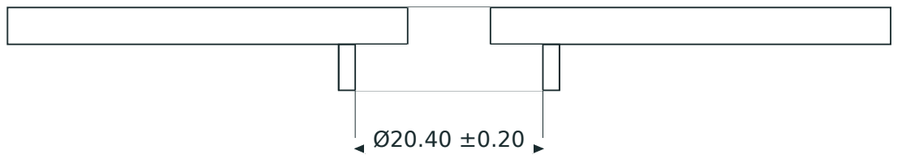
TOP / X–Y



4X R3 OUTER PLATE CORNERS; ARMS AND STOP LUG EXCEPTED

SECTION A–A / PILOT DETAIL

Section through Y=0; arms/horn holes omitted for clarity.



Pilot ring concentric to plate center X=0, Y=0.

Finish OD to Ø24.00 ±0.10; mating lid bore Ø24.40 +0.20/−0.

Minimum nominal-dimension radial clearance: 0.15 mm.

⌖	Ø0.30	A	B	C

Pilot outer axis: basic X=48 from B, Y=20 from C.

F01	**Plate**	96 × 40 × 4; X±48, Y±20, Z34–38; R3 corners
F02	**Arms / pads**	X−48…−44 and44…48; stem Y±6,Z38…88; padR14@Y0,Z88
F03	**Pilot / access**	Ø24 OD / Ø20.4 ID ×5 fromZ29…34; Ø9 access through plate
F04	**Pitch axes**	LEFT Ø4.2; RIGHT Ø6; basicY0,Z88; axis alongX
F05	**Horn fasteners**	2×Ø2.2 through RIGHT at(Y−7,Z88),(Y+7,Z88)
F06	**Yaw horn fasteners**	2×Ø2.2 through plate at(X−7,Y0),(X+7,Y0)
F07	**Yaw stop**	Bridge X45…55,Y±2,Z34…38; Ø4 pin atX55,Y0,Z34…38
F08	**Pitch stops**	2×Ø3 pins axisX, X−45.5…−44.0; centers below
F08a	**Lower stop center**	Y+6.773, Z97.906; basic toA:Z63.906; toC:Y26.773
F08b	**Upper stop center**	Y−8.390, Z96.579; basic toA:Z62.579; toC:Y11.610
F09	**Left inner recess**	Annular Ø28/Ø19 ×1.40 deep; X−45.4…−44; preserve pivot land within Ø19

NOTES: All coordinates in feature schedule refer to CAD assembly axes (mm): X0/Y0 at yaw axis; plate underside Z34.
Prototype process: PETG FFF, 0.2 layers, five perimeters, local solid pivot/root regions; inspect process coupon first.
Deburr and finish-drill critical holes. No supports or burrs on datum A / thrust track. Break exposed edges 0.3 max.
Fit Ø4 sleeve without binding; sleeve + fastener sets pivot endfloat. Do not clamp rotating head. Match drill horn if required.
Integral stop contacts are fault arrestors. Software limits −15°/+25° pitch, ±30° yaw; hard pitch −20°/+30°, yaw ±35°.
General linear tolerances do not apply to boxed basic locations. Inspect hole axes in A|B|C datum frame; GD&T RFS.

NODI-P03 PAN–TILT YOKE MATERIAL: PETG, FFF + FINISH DRILL / PROCESS NOTES ON SHEET 2	**REV A / 2026-10-24** Scale: 1.5:1 views (A3)	**David Borsky** AI-assisted design study
UNITS: mm | THIRD ANGLE PROJECTION | DO NOT SCALE DRAWING General ±0.25 linear / ±1° angular unless limits, basic dimensions or GD&T apply.	ASME Y14.5-2018 principles Unbuilt / inspect before assembly	**SHEET 2 OF 2**